Customer Churn Analysis & Business Intelligence Dashboard

End-to-End Customer Retention, Revenue & Churn Analytics

Engineered an end-to-end customer churn analytics solution by integrating customer, subscription, and support data across three relational tables. Performed data cleaning, feature engineering, KPI development, and business intelligence reporting to identify churn drivers, quantify revenue risk, and build an interactive Tableau dashboard for executive decision-making.

PYTHON | SQL | SQLITE | PANDAS | TABLEAU | MATPLOTLIB | SEABORN

Anurag Patel · Customer Churn Analytics Case Study .

BUSINESS PROBLEM

Business Problem & Analytics Workflow

Why Churn Matters

Customer churn is one of the biggest challenges for subscription-based businesses because it directly impacts recurring revenue, customer lifetime value (CLTV), and long-term profitability. Without identifying high-risk customers early, businesses lose valuable customers and experience unnecessary revenue leakage.

Project Objectives

- Analyze customer churn using relational customer, subscription, and support data
- Identify high-risk customer segments
- Quantify revenue loss and retention opportunities
- Build an interactive Tableau dashboard for business decision-making

End-to-End Analytics Workflow

SQLite Database

Customer, Subscription & Support tables

Python & Pandas

Data merging using Pandas

Data Cleaning

Missing values, standardization, feature engineering

KPI Analytics

20+ KPIs including Churn Rate, ARPU & CLTV

Tableau Dashboard

Interactive business intelligence dashboard

521

Total Customers

Unique customer records in the analytical dataset

3

Relational Tables

Customer, Subscription & Support tables integrated

20+

Business KPIs

Key performance indicators engineered for analysis

1

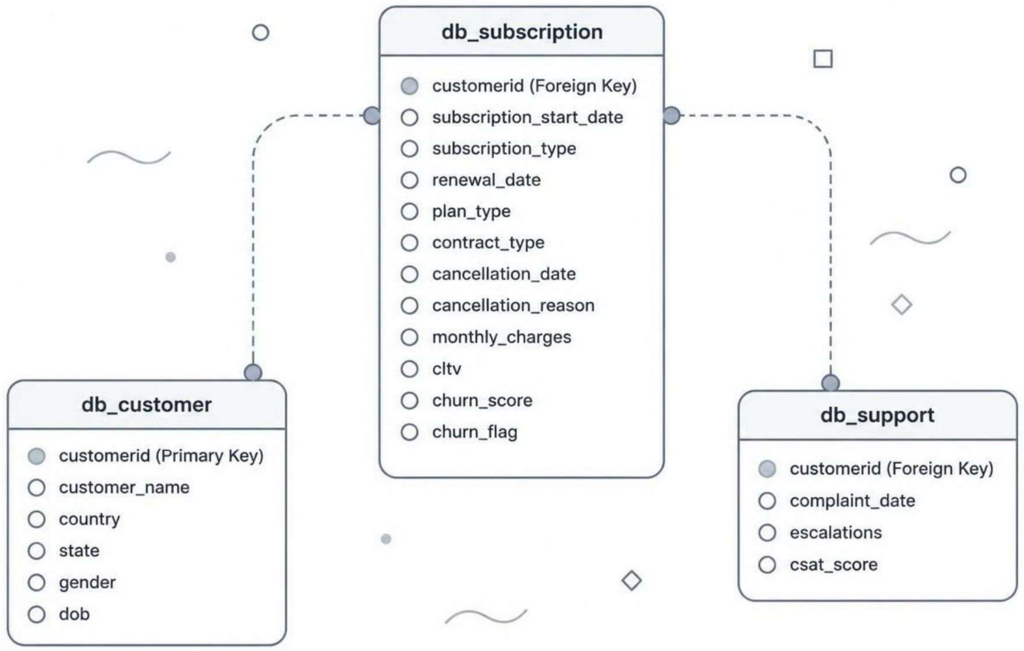
Tableau Dashboard

Interactive executive business intelligence report

DATA ARCHITECTURE

Relational Database Architecture

The customer churn database was designed using a relational data model consisting of three interconnected tables: **Customer**, **Subscription**, and **Support**. Each table is linked through the `customerid` primary key, enabling unified customer-level analytics after data integration. The final analytical dataset contains **521 unique customers**.



521 Customer Records	3 Relational Tables
19 Analytical Columns	customerid Primary Join Key

All three tables were merged using `LEFT JOIN` in Pandas to preserve customer records while enriching subscription and support information for downstream business analytics.

DATA ENGINEERING

Data Preparation & KPI Engineering

Data Cleaning Pipeline

- → Standardized inconsistent gender values (`Man → Male`, `Woman → Female`)
- → Converted DOB and all subscription/cancellation columns to Date format
- → Removed unnecessary columns (`interest`, `pincode`, `col_1`, `comment`)
- → Filled missing country values using existing customer information
- → Merged Customer, Subscription & Support tables using `customerid`
- → Created analytical features: `tenure_days`, `complaint_count`, `churn_flag`, and `churn_risk`

Feature Engineering

Tenure Days
Customer lifetime in days

Complaint Count
Total complaints raised by each customer

Churn Flag
0 = Active, 1 = Churned

Churn Risk
Low / Medium / High customer segmentation

Business KPI Formulas

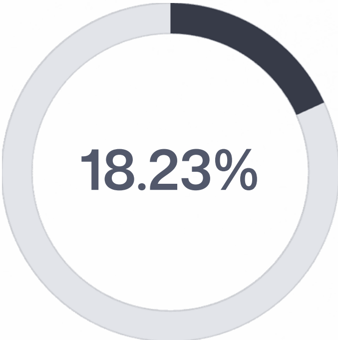
KPI	Formula
Churn Rate	Churned Customers ÷ Total Customers
Retention Rate	1 − Churn Rate
ARPU	Total Revenue ÷ Total Customers
Average Tenure	AVG(Tenure Days)
Revenue at Risk	SUM(Monthly Charges) where Churn Flag = 1
Escalation Rate	Escalated Complaints ÷ Total Complaints
Avg Complaints	Total Complaints ÷ Unique Customers
CLTV	Existing subscription lifetime value metric

Engineering Outcome: Integrated 3 relational tables · Built a unified analytical dataset of 521 customers · Generated 20+ business KPIs · Prepared the dataset for interactive Tableau Business Intelligence

BUSINESS INSIGHTS

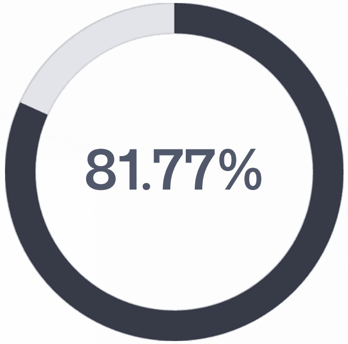
Customer Retention & Revenue Overview

Based on the analysis of **521 customer records**, the following key insights were identified from the customer churn dataset. Approximately **1 in every 5 customers** has churned, while the majority remain active — yet a significant portion of recurring revenue is exposed through churned customers.



Churn Rate

Customers who have cancelled their subscription



Retention Rate

Customers who remain active and subscribed

₹6,764

Total Revenue

₹6,764.79 across all active subscriptions

₹1,254

Revenue at Risk

₹1,254.05 exposed through churned customers

₹12.98

ARPU

Average Revenue Per User per month

1,446

Avg Tenure (Days)

Average customer lifetime in days

REGIONAL & SUBSCRIPTION ANALYSIS

Regional Churn & Subscription Performance

Regional Churn Analysis

Meghalaya recorded the highest churn rate at **66.67%**, making it the highest-risk region by a significant margin. This state requires immediate investigation due to its unusually high customer attrition compared to all other regions in the dataset.

⚠ Meghalaya's 66.67% churn rate is an outlier that demands urgent regional investigation — potential drivers include pricing sensitivity, service quality gaps, or localized competitive pressure.

Subscription Plan Performance

The **Premium Plan** recorded the highest churn rate at **20.47%**. Since Premium customers generally contribute higher revenue per user, this churn has a disproportionately greater financial impact than cancellations in lower-tier plans.

- Premium customers represent the highest-value segment
- Higher churn in this tier amplifies revenue loss
- Renewal pricing and feature adoption require review
- Loyalty discounts or personalized renewal offers recommended

TREND & SUPPORT ANALYSIS

Monthly Churn Trend & Customer Support Insights

Monthly Churn Trend

The highest number of customer cancellations occurred in **September 2024**, indicating a possible business event such as pricing changes, service issues, or product updates during that period. This spike warrants a detailed root-cause review across product, billing, and customer success teams.

Review product releases, subscription price changes, payment failures, marketing campaigns, and service outages that occurred during September 2024.

Customer Support Analysis

Nearly **half of all customer complaints** required escalation, indicating significant opportunities to improve first-level support efficiency and reduce operational overhead.

50.48%	0.01
Escalation Rate	Escalation–Churn Correlation

Escalations alone show almost no linear relationship with churn, suggesting that churn is influenced by **multiple factors** rather than support interactions alone.

STRATEGIC RECOMMENDATIONS

Action Items & Strategic Priorities

Investigate Meghalaya

Analyze whether high churn is driven by pricing, technical issues, poor service quality, or customer complaints. Conduct regional customer feedback surveys to identify root causes.

Review the Premium Subscription Plan

Examine renewal pricing, feature adoption, and cancellation reasons. Introduce loyalty discounts or personalized renewal offers for Premium customers to reduce the 20.47% churn rate.

Analyze the September 2024 Churn Spike

Review product releases, subscription price changes, payment failures, marketing campaigns, and service outages that occurred during September 2024.

Protect ₹1,254.05 Revenue at Risk

Prioritize high-CLTV customers approaching renewal through targeted email, SMS, and customer success outreach campaigns to minimize recurring revenue leakage.

Reduce Support Escalations

Improve first-contact resolution, monitor low CSAT tickets, and establish faster response SLAs to decrease the 50.48% escalation rate and improve customer satisfaction.

Build a Multi-Factor Retention Strategy

Since escalation and churn have almost no correlation (0.01), combine churn score, tenure, CLTV, subscription plan, and CSAT to identify customers at the highest risk of leaving.

EXECUTIVE CONCLUSION

Executive Summary

This project demonstrates a complete **Data Analytics & Business Intelligence workflow** — from raw relational data to executive reporting — showcasing practical skills in data engineering, KPI analysis, visualization, and business decision support suitable for real-world customer retention use cases.

Project Overview

Engineered an end-to-end Customer Churn Analytics solution by integrating customer, subscription, and support data from three relational SQLite tables. Built a complete analytics pipeline covering data cleaning, feature engineering, KPI development, exploratory data analysis, and interactive Tableau dashboard creation.

Business Impact

Analyzed 521 customer records across three relational datasets. Generated 20+ business KPIs including Churn Rate, Retention Rate, ARPU, Revenue at Risk, CLTV, Tenure, CSAT, and Escalation Rate. Identified high-risk customer segments and quantified revenue exposure through churn analysis.

Technical Skills Demonstrated

Programming: Python, SQL, Pandas, NumPy · **Database:** SQLite, Relational Data Modeling · **Business Intelligence:** Tableau, Dashboard Design, KPI Engineering · **Analytics:** Customer Segmentation, Churn Analysis, Revenue Analytics, CLTV, Data Storytelling

> The business should prioritize reducing Premium plan churn, investigate Meghalaya's exceptionally high churn rate, address the September 2024 cancellation spike, and proactively retain high-value customers to minimize revenue loss.